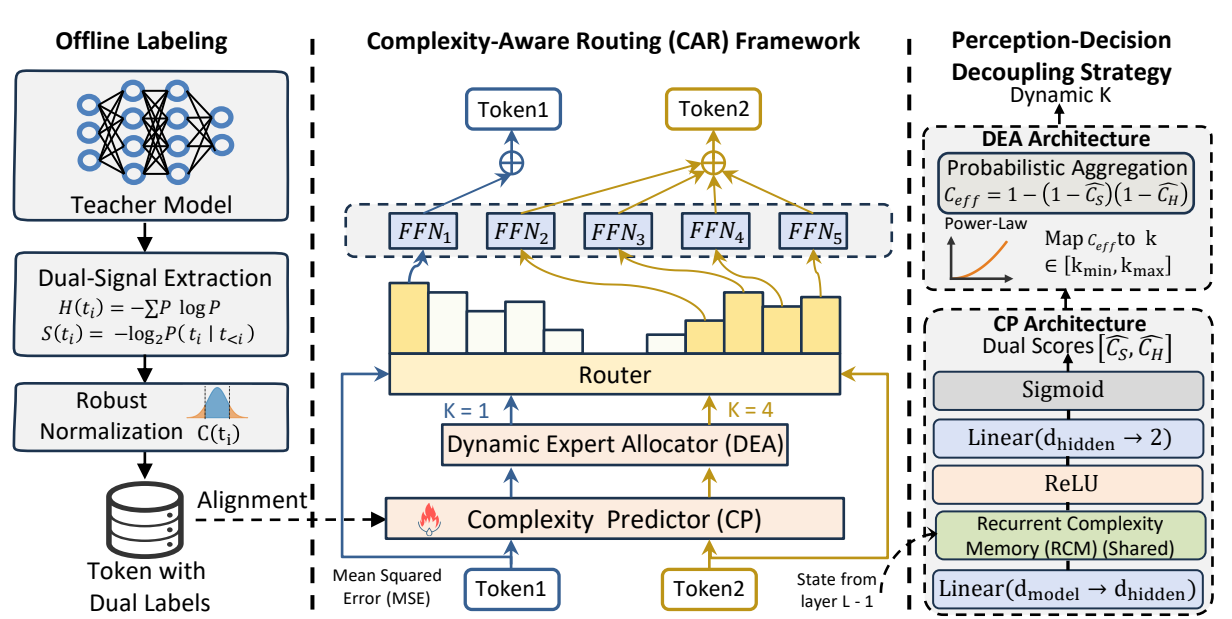

Offline Labeling
Teacher Model
Dual-Signal Extraction
$H(t_i) = -\sum P \log P$
$S(t_i) = -\log_2 P(t_i \mid t_{<i})$
Robust Normalization
$C(t_i)$
Alignment
Token with Dual Labels
Complexity-Aware Routing (CAR) Framework
Token1
Token2
FFN_1
FFN_2
FFN_3
FFN_4
FFN_5
Router
K = 1
K = 4
Dynamic Expert Allocator (DEA)
Complexity Predictor (CP)
Mean Squared Error (MSE)
Token1
Token2
State from layer L - 1
Perception-Decision Decoupling Strategy
Dynamic K
DEA Architecture
Probabilistic Aggregation
$C_{eff} = 1 - (1 - \widehat{C_S})(1 - \widehat{C_H})$
Power-Law
Map c_{eff} to k $\in [k_{min}, k_{max}]$
CP Architecture
Dual Scores $[\widehat{C_S}, \widehat{C_H}]$
Sigmoid
Linear($d_{hidden} \rightarrow 2$)
ReLU
Recurrent Complexity Memory (RCM) (Shared)
Linear($d_{model} \rightarrow d_{hidden}$)